

MODULES 11 TO 19 MASTER EXAM PREP GUIDE

OFFICIAL CURRICULUM REVIEW

Cisco Certified CyberOps Associate (CBROPS 200-201)

Scope: Chapters 11, 12, 13, 14, 15, 16, 17, 18, 19
Format: High-Yield Knowledge & Exam Traps

1. KEY COMPARISON MATRICES (THE TOP MOST-TESTED CONCEPTS)

TACACS+ vs. RADIUS (Module 19)

Feature	TACACS+	RADIUS
Protocol / Port	TCP Port 49	UDP Ports 1812 / 1813
Encryption	Entire packet payload	Password field only
AAA Separation	Strictly separates Auth/Authz	Combines Auth/Authz
Primary Role	Device Administration	Network Access (802.1X/VPN)

SPAN vs. Network TAP (Module 15)

Feature	SPAN (Port Mirror)	Network TAP
Nature	Switch software feature	Dedicated physical hardware
Switch Impact	Consumes CPU; drops frames	Zero load on network switches
Corrupted Packets	Drops CRC/runt frames	Captures 100% of malformed frames
Power Needed	Powered via switch	Passive fiber TAPs require no power

IPsec Protocols & Modes (Module 12)

Protocol / Mode	Confidentiality (Encryption)	Integrity / Auth	IP Header Behavior
AH (Protocol 51)	NO (Cleartext payload)	YES	Protects original IP header
ESP (Protocol 50)	YES (AES/3DES)	YES	Protects payload + ESP trailer
Transport Mode	Encrypted payload only	YES	Keeps original IP header
Tunnel Mode	Encrypted original packet	YES	Adds a NEW outer IP header

Access Control Models (Module 19)

Model	Decision Basis	Authority
DAC	Resource Owner's discretion	Data Creator (NTFS, chmod)
MAC	Sensitivity labels vs. clearance	Central OS / Military (SELinux)
RBAC	Organizational job role	System Administrator (AD Groups)
ABAC	Dynamic attributes (time, device)	Policy Engine (Cloud IAM, ISE)

2. LAYER-BY-LAYER ATTACK & DEFENSE SUMMARY (MODULES 16 & 17)

Protocol Layer	Primary Attack Vectors	Mechanics / Indicators	Defensive Countermeasure
Layer 2: ARP	ARP Poisoning / Spoofing	Forged Gratuitous ARP replies corrupt local ARP caches, creating a Man-in-the-Middle (MitM).	Dynamic ARP Inspection (DAI) with DHCP Snooping binding table.
Layer 2: DHCP	DHCP Starvation & Rogue DHCP	Flooding fake MACs (Yersinia) exhausts IP pool; rogue DHCP server assigns fake gateway and DNS.	DHCP Snooping (Untrusted ports drop incoming DHCP offers).
Layer 3: IP	Teardrop & IP Spoofing	Overlapping fragment offset fields crash reassembly buffers; spoofed source IPs bypass filters.	Drop overlapping fragments; egress/ingress anti-spoofing ACLs (RFC 2827).
Layer 3: ICMP	Smurf & Ping of Death	Smurf sends spoofed ping to directed broadcast; Ping of Death exceeds 65,535-byte IP size.	no ip directed-broadcast on Cisco router interfaces.
Layer 4: TCP	SYN Flood & TCP Reset (RST)	Exhausts half-open connection backlog table; forged RST packets terminate active sessions.	TCP SYN cookies, stateful inspection, TCP sequence number randomization.
Layer 7: DNS	DNS Tunneling & Cache Poisoning	Exfiltrates data inside subdomain queries (exfil.attacker.com); injects fake IP mappings.	DNSSEC cryptographic signing; Cisco Umbrella DNS filtering.

Protocol Layer	Primary Attack Vectors	Mechanics / Indicators	Defensive Countermeasure
Layer 7: Web	SQLi, XSS, and CSRF	SQLi extracts database records; XSS executes client-side scripts; CSRF hijacks user state.	Parameterized Queries , input validation, Anti-CSRF synchronizer tokens.

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(PART 2)

Cisco Certified CyberOps Associate • Operations, Metrics & Architecture

SOC OPERATIONS &
PROTOCOLS

3. SOC OPERATIONS, METRICS & DISASTER RECOVERY (MODULE 18)

SOC ALERT MATRIX The 4 Alert Classifications

- **True Positive (TP):** Malicious attack occurring \$ ightarrow\$ Alert generated. *(Accurate detection)*
- **False Positive (FP):** Benign business traffic \$ ightarrow\$ Alert generated. *(Nuisance alert / noise)*
- **True Negative (TN):** Normal traffic \$ ightarrow\$ No alert generated. *(Proper operation)*
- **False Negative (FN):** Malicious attack occurring \$ ightarrow\$ **NO alert generated.** *(Worst-case catastrophic failure)*

METRICS & DRP Disaster Recovery Objectives

- **RTO (Recovery Time Objective):** Maximum tolerable **downtime duration** before business catastrophe.
- **RPO (Recovery Point Objective):** Maximum tolerable **data loss measured in time.**
- **Hot Site:** Real-time replicated data and live hardware; failover in **minutes/hours.**
- **Warm Site:** Hardware present; software/data must be loaded; failover in **days.**
- **Cold Site:** Empty facility with power/HVAC; hardware must be ordered; failover in **weeks.**

4. HIGH-YIELD MUST-KNOW PORT NUMBERS & PROTOCOL CHECKLIST

Port Number	Protocol Name	Transport	Exam Relevance & Trap Warning
22	SSH	TCP	Encrypted terminal management. Replaces unencrypted Telnet (Port 23).
49	TACACS+	TCP	Cisco AAA device admin. Encrypts ENTIRE payload , separates authentication and authorization.
53	DNS	UDP / TCP	UDP for regular queries (<512 bytes); TCP for Zone Transfers (AXFR) and responses >512 bytes.
67 / 68	DHCP	UDP	Port 67 = Server; Port 68 = Client. Targeted by starvation and rogue server attacks.
123	NTP	UDP	Network Time Protocol. Crucial for forensic log correlation; abused in reflection DDoS attacks.
514	Syslog	UDP	System logging messages. Severity levels range from 0 (Emergency) to 7 (Debugging).
1812 / 1813	RADIUS	UDP	Port 1812 = Auth; Port 1813 = Acct. Encrypts ONLY password ; combines auth and authorization.
IP Proto 50	IPsec ESP	Layer 3	Encapsulating Security Payload: provides encryption , authentication, and integrity.
IP Proto 51	IPsec AH	Layer 3	Authentication Header: provides authentication and integrity ONLY (NO encryption).

5. SWITCH PORT SECURITY & 802.1X ACCESS CONTROL (MODULE 19)

Port Security Violation Modes

- **Protect:** Drops unauthorized frames. Does NOT send syslog or increment violation counter.
- **Restrict:** Drops unauthorized frames. Sends a syslog message and increments violation counter.
- **Shutdown (Default):** Drops frames, sends syslog, and disables port into err-disable.
- **Sticky MAC:** Dynamically learned MAC address automatically written to running-config.

802.1X Architecture Triad

- **Supplicant:** The client software on the end device requesting network access.
- **Authenticator:** The intermediary switch or Wireless LAN Controller (WLC) enforcing the port block.
- **Authentication Server:** The centralized database (Cisco ISE / RADIUS) verifying user identity.
- **EAP-TLS:** Requires certificates on **both client and server.**